



Silesian
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of Technology

MASTER THESIS

Applying AI Algorithms to Train a Computer Player in a Logic Board Game

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Thesis title

Applying AI Algorithms to Train a Computer Player in a Logic Board Game

Abstract

(Thesis abstract – to be copied into an appropriate field during an electronic submission – in English.)

Key words

(2-5 keywords, separated by commas)

Tytuł pracy

Zastosowanie algorytmów sztucznej inteligencji do nauki komputerowego gracza w logicznej grze planszowej

Streszczenie

(Streszczenie pracy – odpowiednie pole w systemie APD powinno zawierać kopię tego streszczenia.)

Słowa kluczowe

(2-5 słów (fraz) kluczowych, oddzielonych przecinkami)

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Chapter 1

Introduction

Board games have entertained people since ancient times, from teaching techniques for the youngest, to providing entertainment for entire families, to competitions for the best players. Nowadays, many popular board games have been introduced into the digital world to bring people together from different corners of the globe. The digital board game environment has also made it possible to play games without the participation of physical people on the other side of the screen. The implemented players were tasked with behaving like human participants.

This thesis focuses on applying AI algorithms to train a computer player for a logic-based board game. Unlike traditional games such as chess, the selected game introduces distinct challenges, including combinatorial complexity, dynamic board states, randomness and the necessity for both tactical and strategic reasoning. Designing an AI capable of playing effectively in such an environment requires balancing rule-based knowledge with adaptive learning techniques

1.1 Introduction into the problem domain

The history of board games spans thousands of years, reflecting their enduring role in human culture. The Royal Game of Ur, originating in ancient Mesopotamia more than 4,500 years ago, is considered one of the earliest known board games [1]. Since then, board games have remained a popular pastime across civilizations, valued not only as a form of entertainment but also as a means of social interaction and intellectual challenge.

Beyond recreation, board games have also been recognised for their pedagogical value. Recent studies highlight their efficacy as educational tools, showing that board games can enhance motivation, engagement, and learning outcomes in classroom settings [3]. By providing structured environments that require logical thinking, planning, and decision-making, board games naturally support the development of cognitive and social skills.

Board games have long been used as a way to test logical thinking and strategy. Unlike purely abstract games such as Chess, many modern tactical board games introduce an

element of randomness, such as shuffled decks, dice rolls, or random tile draws. This combination of determinism and uncertainty contributes to a more complex decision-making process. Players are required to plan tactics based on the current position, whilst also having the capacity to adapt to unpredictable events.

The dual role of board games as both an entertainment medium and an educational tool renders them an ideal domain for exploring artificial intelligence. These games combine structured rule systems with varying levels of complexity, randomness, and strategic depth. Consequently, the training of a computer player for such games contributes not only to the field of game AI but also offers broader insights into decision-making processes, learning under constraints, and adaptive strategies in uncertain environments.

1.2 Settling of the problem in the domain

Many popular board games that use artificial intelligence algorithms only required learning the rules and adapting tactics to the opponent. One such game is chess, which has many artificial intelligence algorithms implemented, one of which is **Stockfish - Open Source Chess Engine** [2]. However, in numerous games, players are required to adjust their tactics not only to their opponents' actions, but also to environmental and situational randomness. Rummikub exemplifies this, requiring players to adapt their strategy to other players, the random draw of tiles, and the specific rules for laying down sets.

In Rummikub, players create and rearrange sets and sequences of numbered tiles in different colors. The objective of the game is to place all tiles from one's hand onto the shared board by forming valid combinations according to the game's rules. However, in contrast to purely deterministic games, Rummikub introduces stochastic elements, such as random tile draws or randomly generated tiles at the beginning of the game. These elements have been shown to significantly affect the course of the game. This randomness forces both human and artificial players to adapt their tactics dynamically to ever-changing conditions on the board.

From an AI perspective, this environment poses several challenges. As the board evolves, the number of potential moves increases, rendering an exhaustive search impractical. The uncertainty of future tile draws and opponent actions introduce an additional layer of complexity that requires probabilistic reasoning or learning-based adaptation rather than deterministic planning alone.

Positioning this problem within the broader field of game AI highlights its research value. Developing an effective computer player for Rummikub requires combining strategic evaluation, pattern recognition, and adaptive learning. It also provides an opportunity to investigate how modern AI algorithms, such as reinforcement learning and simulation-based decision-making, can operate effectively in environments that are logical in structure but influenced by randomness.

1.3 Objective of the thesis

The main objective of this thesis is to design, implement, and train an artificial intelligence agent capable of autonomously playing the board game Rummikub in an effective and strategic way. The developed AI system should be able to analyze the current game state, evaluate available legal moves, and select actions that maximize its probability of success in a random environment.

Due to the stochastic nature of Rummikub, particularly the randomness associated with tile distribution and draws, the agent must be able to operate in a dynamic and partially unpredictable environment. Therefore, a key objective of this work is to develop an AI that can adapt its strategy to changing board configurations, varying hand compositions and evolving tactical opportunities throughout the game.

1.4 Scope of the thesis

This thesis focuses on the application of Reinforced Learning algorithms to train a computer player for the board game Rummikub within a custom-developed digital environment. The scope of the work includes the design and implementation of the game logic, the integration of AI decision-making mechanisms, and the experimental evaluation of the trained agent. The project is limited to a single board game, Rummikub, and does not attempt to generalise the developed solution to other game genres. The research excludes online multiplayer deployment and human-subject studies. The implementation is carried out using the Unity engine, where the game environment and AI agent are developed and tested.

The thesis maintains focus on the core research problem: the practical application of AI algorithms to enable intelligent decision-making in a tactical board game environment characterised by combinatorial complexity and stochastic elements. The development of the project include:

- Implement game environment for the player - development of a digital representation of the Rummikub board game using the Unity engine, including tile management, valid set detection, and interaction rules.
- Integrating artificial intelligence - implementation of the different reinforcement learning algorithms available from Unity ML-Agents Toolkit.
- Developing a system of rewards and penalties - implementation of properly working points system for better training the model.
- Training the agent - training of the computer player based on simulated games, allowing it to improve its tactics through experience and feedback.

- Evaluating the model - conduction of the experiments to evaluate the efficiency, adaptability and success rate of the trained AI.

1.5 Short description of chapters

TODO at the end of whole thesis

1.6 Clear description of contribution of the thesis's author

The essential elements of the project, that include the author contributions, are:

- Development of experiments environment - the design and implementation of a fully functional digital Rummikub environment, including game logic, rule validation, and agent interaction mechanisms.
- Supervision and configuration of the learning process – the definition of the state and action spaces, design of the reward structure, configuration of training parameters, and monitoring of the agent's learning progress.
- Evaluation of the agent's performance – the design and execution of simulation-based experiments to assess the effectiveness, adaptability, and decision-making quality of the trained AI model.
- Analysis and interpretation of results – the examination of experimental outcomes and formulation of conclusions regarding the applicability of artificial intelligence methods to tactical board games with stochastic elements.

Chapter 2

Analysis of the problem

References to book [4], scientific papers in journals [5], conference papers [6], and web pages [7].

Equations should be numbered:

$$y = \frac{\partial x}{\partial t} \tag{2.1}$$

Chapter 3

Subject of the thesis

tekst

3.1 [Section title]

3.2 [Subsection title]

Each figure in the document should be referred to at least once (fig. 3.1).

Each table in the document should be referred to at least once (Tab. 3.1).

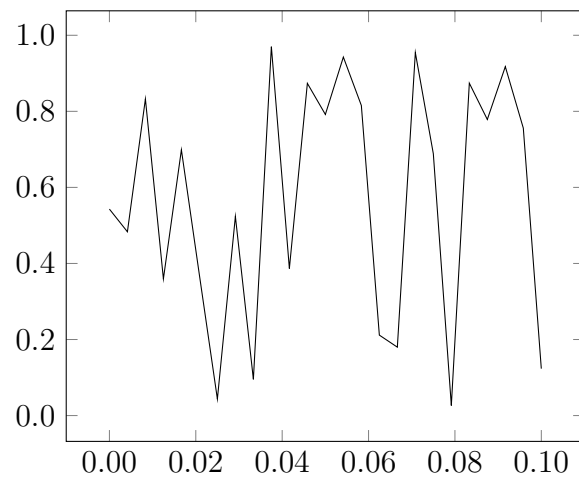


Figure 3.1: Figure caption.

Table 3.1: A caption of a table is ABOVE it.

ζ	method						
	alg. 1	alg. 2	alg. 3			alg. 4, $\gamma = 2$	
			$\alpha = 1.5$	$\alpha = 2$	$\alpha = 3$	$\beta = 0.1$	$\beta = -0.1$
0	8.3250	1.45305	7.5791	14.8517	20.0028	1.16396	1.1365
5	0.6111	2.27126	6.9952	13.8560	18.6064	1.18659	1.1630
10	11.6126	2.69218	6.2520	12.5202	16.8278	1.23180	1.2045
15	0.5665	2.95046	5.7753	11.4588	15.4837	1.25131	1.2614
20	15.8728	3.07225	5.3071	10.3935	13.8738	1.25307	1.2217
25	0.9791	3.19034	5.4575	9.9533	13.0721	1.27104	1.2640
30	2.0228	3.27474	5.7461	9.7164	12.2637	1.33404	1.3209
35	13.4210	3.36086	6.6735	10.0442	12.0270	1.35385	1.3059
40	13.2226	3.36420	7.7248	10.4495	12.0379	1.34919	1.2768
45	12.8445	3.47436	8.5539	10.8552	12.2773	1.42303	1.4362
50	12.9245	3.58228	9.2702	11.2183	12.3990	1.40922	1.3724

Chapter 4

Experiments

Chapter 5

Summary

- synthetic description of performed work
- conclusions
- future development, potential future research
- Has the objective been reached?

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Appendices

Technical documentation

List of abbreviations and symbols

DNA deoxyribonucleic acid

MVC model–view–controller

N cardinality of data set

μ membership function of a fuzzy set

$\mathbb{E}$ set of edges of a graph

$\mathcal{L}$ Laplace transformation

List of additional files in electronic submission (if applicable)

Additional files uploaded to the system include:

- source code of the application,
- test data,
- a video file showing how software or hardware developed for thesis is used,
- etc.

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